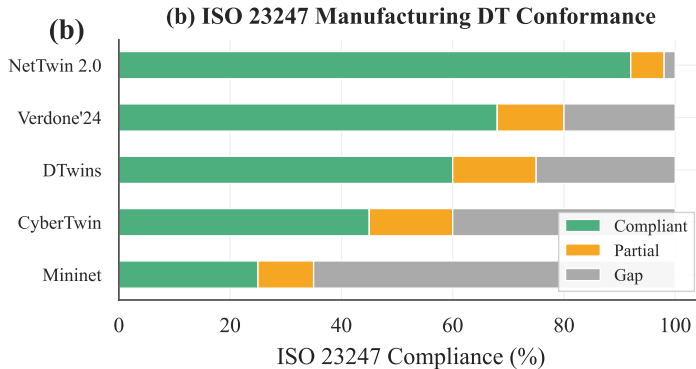


State-of-the-Art Digital Twin Literature Against Standard Conformal Digital Twins

(a)

Digital Twin Framework Venue		Live Sync	Conformal Cal.	Causal RCA	Auto Response	Safety Gates	Scale (Nodes)
Verdone et al.	IEEE TNSM'24	Yes (Periodic)	No	No	Heuristic	No	120
DTwins	ACM SoCC'23	Yes (Trace)	No	Correlation	No	No	250
CyberTwin	IEEE TDSC'22	No (Offline)	No	Bayesian	Rule-based	Partial	80
Azure Digital Twins	Commercial	Yes (PubSub)	No	No	Manual	No	500
GNS3 / Mininet	Open Source	Emulation	No	No	No	No	150
NetTwin 2.0 (Ours)	IEEE TNSM	Yes (<2ms)	Yes (ACI)	Yes (CausalNet)	Yes (Bandit)	Yes (Formal)	2,000

(c) Multi-Dimensional Performance Radar



(c)

